



RAMCO INSTITUTE OF TECHNOLOGY

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NBA Accredited UG Programs: CSE, EEE, ECE and MECH

Department of Information Technology

Academic Year: 2024 - 2025(Even Semester)

Degree, Semester & Branch: B. Tech, VI & IT

Course Code & Title: CCS335 & Cloud Computing

Name of the Faculty member: A. Alagulakshmi, AP/IT

Innovative Practice Description

- **Unit / Topic:** Unit III/ Docker Components
- **Course Outcome:** CO3
- **Activity Chosen:** Flipped class Room
- **Justification:**

The flipped classroom model promotes active and self-directed learning. In the approach, students review instructional materials such as video lectures and notes at home, allowing in class time to be utilized for discussion, presentations, and collaborative tasks. This enhances students understanding, promotes self-learning, and encourages deeper classroom engagement.

Time Allotted for the Activity: 15 Minutes

Details of Implementation:

The flipped classroom approach was implemented in three structured phases to enhance student engagement and understanding. In the Content Sharing phase, students were provided with relevant study materials such as lecture videos, notes, and reference links on Docker Components through online platforms ahead of the scheduled class. This was followed by the Self-Learning Phase, where students were encouraged to independently explore and study the shared content to develop a foundational grasp of Docker and its various components. The final Classroom Activity phase involved active student participation through presentations and a short assessment to reinforce their learning. During this session, Ms. Aarthi G presented her insights and understanding of Docker Components, showcasing the knowledge she had gained through self-learning. Her presentation is depicted in Figure 1, highlighting the effectiveness of the flipped classroom model in promoting self-directed learning and peer knowledge-sharing.

CO – PO / PSO mapping:

CO	PO1	PO2	PO5	PO10	PO11	PO12
CO3	3	2	2	2	2	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1 3	PO2 2	PO3 2	PO10 2	PO11 2	PO12 2
Justification for correlation	Enables students to apply IT concepts to manage Docker-based virtualization	Encourages exploration of Docker tools and containerization techniques.	Supports the evaluation of constraints and tools relevant to software deployment.	Enhances analytical skills in understanding technical documentation and presentations.	Develops the ability to assess the application of Docker in real-time engineering environments.	Fosters continuous learning through self-exploration of emerging technologies in cloud computing.

- **Images / Screenshot of the practice:**



Fig 1: Aarthi G presentation about Docker Components

Reflective Critique:

- ❖ **Feedback of practice from students and other stakeholders:**

- Students appreciated the flexibility and self-paced nature of learning.
- They expressed confidence in exploring complex topics independently before classroom discussion.

- ❖ **Benefit of the practice:**

- Increased one-on-one interaction between faculty and students.
- Encouraged collaborative learning and peer discussions.
- Allowed students to learn at their own pace.
- Reduced the impact of missed classes.
- Motivated students to come prepared for classroom sessions.

❖ ***Challenges faced in implementation:***

- The depth of understanding varied across individuals and groups.
- Developing the flipped classroom materials required significant time and effort from the faculty.

References:

- ❖ <https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model>
- ❖ https://en.wikipedia.org/wiki/Flipped_classroom
- ❖ [https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162- what-why-and-how-to-implement-a-flipped-classroom-model](https://omerad.msu.edu/teaching/teaching-skills-strategies/27-teaching/162-what-why-and-how-to-implement-a-flipped-classroom-model)

Signature of Faculty Member

HOD

